Industrial Safety Processes

Hoisting &
Conveying

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Hoist Basics

- The rated load capacity must be indicated on *the body* of the machine
  - Should not be buried in a “user’s manual”
  - Should be not indicated on a sign on the wall
- This load rating is applicable to any device used to hoist or its supporting structure
  - Lifting beam in elevator machine room

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Basic In-Service Hoists

- Inspection and maintenance is critical
- Failures of hoists result in serious incidents on a regular basis
- All components need to be maintained according to manufacturer specifications
  - Operating environment is a major consideration as well

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Electric Hoists

- Typically a wire rope (cable) system
- May or may not include rope redundancies
- Inspection and maintenance of ropes and sheaves very important
- Limit switches should be installed to ensure that enough rope remains on the drums

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Chain and Air Hoists

- Usually portable units
- May be fixed to a wall for regular use
  - Common in some chemical environments
  - Over floor pits for inter-floor transfer
- Common in automotive repair and other maintenance shops
- Usually designed for one-man/woman use

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Limit Devices

- Depending on the hoist, it may come with limit devices
- Use the limit devices as warning signals to stop operations, not to stop loads normally
- Limit operations:
  - Up and down (protect hoist and material)
  - Travel (protect hoist and other objects)
  - Load (protect against overloading)

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Lifting

- Know the weight of the load
- Never lift over people
  - May be allowed under OH&S Code under special circumstances
- Always lift from immediately above the load
  - Avoid swinging motion on ropes/lines
  - Avoid hitting people/objects
  - Swing and angles create additional load on the ropes/lines – may cause failure

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Setup & Drum Issues

- Make sure sufficient rope/wire is on the drum at all times
  - Check MOM – usually at least 2 wraps minimum
- Ensure that rope/wire is wound on drum so that it doesn't damage the line
- Initial care and attention can save in wear and tear – and incidents

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Inspection & Maintenance

- Always follow manufacturer’s instructions
- Key elements to inspect are:
  - Hooks
  - Ropes
  - Sheaves
  - Drums

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Inspection of Hooks

- Hooks should be clean and free from cracks and defects
- Bend from plane cannot exceed 10%
- Throat openings cannot be stretched more than 15% of normal size (8% for latched hooks)

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Design & Construction

- Several different designs
- Design matches function
- Design matches location
- Need to consider environment (indoor/outdoor)
- Need to consider operation and maintenance costs
  - Need qualified operator or not?

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Guards & Limit Devices

- Guarding of accessible moving parts is a major concern
  - Gears, chains, sprockets, etc.
  - Fouling systems could result in damage and crane failure
- Limit devices should be tested and used as per manufacturer specifications
  - Travel of crane/hoist, overload, upper/lower switches(up/down)

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Limit Devices

- Should be tested at the start of each shift
  - Overload alarms
  - Upper and lower limits
  - Travel limits
- Follow manufacturer's instructions so that equipment isn't damaged
- Don't use limit switches as your stop/starts – could result in failure

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Crane & Hoist Signals

- Hand signals commonly used to assist crane operator
- Standard signals are common, but make sure – never assume
- Operator should only follow the orders of the signaller
  - Special clothing, communication system
- BUT, must follow EMERGENCY STOP signal from anybody

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http://www.mcleodcranes.co.nz/cranes/signals/

Selection & Training of Operators

- Operators should meet physical requirements for the job
- Common for annual physicals and tests similar to truck drivers
- In most provinces, the operator must be “competent”, usually meaning:
  - Registered Apprentice
  - Journeyman
  - Recognized related trade certificate

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Basic Operating Rules

- Never lift over people
  - Unless necessary
  - People are informed/aware of operations
- Lift load straight up – not at angle
  - Avoid swinging into people/objects
  - Avoid creating additional loads on lines
- Do not trip limit switches deliberately other than for manufacturer's required testing

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Types of Cranes

- Overhead
- Gantry
- Derrick
- Jib
- Tower
- Mobile
- Portable

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Main Safety Issues with Cranes

- Improper lifts
- Overloading
- Inadequate inspections and repairs
- Failures are usually catastrophic in nature
  - Fatalities, major losses
- Contact with electrical lines

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Improper Lifts & Overloading

- Attempting loads in excess of rated capacity
  - Mobile cranes and angle of boom
  - Inaccurate estimate of load weight
- Attempting loads with improper rigging
  - Failure to secure loads properly
- Lifting over personnel
- Failing to set mobile crane on solid surface

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Inadequate Inspections & Repairs

- Not checking all required areas
- Failure to maintain proper equipment logs
- Failure to check rigging properly
  - Untrained staff
  - Don’t know what to look for
- Lack of thorough inspections following “near miss” incidents

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Crane Collapses or Flips

- Usually fatal
- Usually result in total loss of crane and cargo
- Often involve secondary damage to adjacent properties, equipment
- Usually required to report to local OHS regulator, even if no injuries
  - Will likely result in shutdown of workplace

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Bad Planning, Rushing & Line of Fire

- Staying out from under loads or “line of fire” still a major issue
- Prepare, plan and take your time
- Knowing weight of load, load limits, and respecting alarms are key

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Contact with Live Electrical

- Major issue with mobile cranes
  - Also with genie/scissor lifts in some areas
- Unless not possible, lines should be deenergized
- Check safe limits of approach in local legislation
- Crane operator can be electrocuted if he/she does not follow proper procedure
- “No hand” dismount and shuffle rather than hop

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Types of Conveyors

- Roller
- Belt
- Slat/apron
- Chain
- Shackle
- Screw
- Bucket
- Portable
- Gravity/vertical, etc.

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Conveyors and Power

- Conveyors may have motor power for all or part of it’s operation
- Power and movement may be controlled centrally or at specific locations (on/off switches)
- Movement could be triggered by presence sensors

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General Precautions

- Guarding of exposed rotating parts
  - Chains, sprockets, gears, etc.
  - Especially in areas easily or regularly accessible
- Controlling speed of operation
  - Ejection of materials, loss of product, striking workers
- Use for materials, not people
- Emergency stopping devices

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